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Editors
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Dear Editors:

Please consider the enclosed Research Report as a Direct Submission to *PNAS*: "Dynamic Arrest Follows Assembly on a Temperature-Scaled Common Clock in an Elastin-like Polypeptide."

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Classification. Physical Sciences: Applied Physical Sciences. Secondary: Biophysics and Computational Biology.

Significance. Whether a soft or biological material has stopped flowing is usually inferred from its structure: a scattering profile that has grown enough is read as a gel. We measure structure and motion on the same detector frames and find that they run on one clock. In isothermal experiments on the sequence-defined elastin-like polypeptide $(\mathrm{VPAVG})_{30}$ at four temperatures between 28.00 and 28.75 °C, every sample assembles, but only those above 28.00 °C arrest within the experiment. The low-Q SAXS intensity and the arrested fraction collapse with the same scaling temperature, and arrest follows assembly: every sample that arrests is half arrested at nearly the same intensity, and most of the arrest happens after most of the low-Q structure has formed. Arrest reaches the largest length scales first. At 175 nm the fast-decay amplitude falls from 0.80 to 0.07 within an hour, while the $1/e$ rate of the fast decay levels off.

Novelty. One rescaling of the waiting time collapses all four isothermal experiments onto a single master curve: with $T_s = 0.24 \pm 0.02$ K, arrest speeds up by about a factor of three for every quarter of a degree of heating, and the low-Q SAXS superposes with the same T_s. That is a stronger statement than a rate: most of what distinguishes the four experiments is their pace, and temperature sets it. A scaling temperature of a few tenths of a kelvin is far too sharp for a single contact, which places the rate-limiting step at the concerted formation of a junction in which many contacts close together.

That scaling framework is our own. One of us introduced it for thermoreversible colloidal gels [Q. Zhang *et al.*, *Phys. Rev. Lett.* **119**, 178006 (2017)], where it returns $T_s = 0.33$ and 0.45 K for two particle sizes [D. Bahadur *et al.*, *J. Chem. Phys.* **151**, 104902 (2019)]. The present work carries it from colloids to a protein polymer and finds the same cooperative regime. We cite the agreement with the nanoparticle-gel phenomenology rather than claiming it as new; what is new is that structure and dynamics share one clock in a protein network, and that one rescaling governs every temperature we can measure.

Why it needed this measurement. The fast relaxation is sub-millisecond at these length scales and falls at or below the shortest delay a conventional kilohertz acquisition reaches. Recording the scattering at 50 kHz with the XSPA-500k at APS 8-ID-I resolves the fast decay itself: its rate, its diffusive Q dependence and its broadening as the sample arrests. At conventional speeds only the loss of contrast would remain.

Limits, stated plainly. The 1.6 s maximum delay bounds a fraction arrested on that scale rather than an infinite-time nonergodicity factor. The scaling rests on three arresting temperatures spanning half a degree, and the collapse is approximate rather than exact.

Companion manuscript. A separate manuscript, in preparation from the same construct, reports the structural side: a thermally reversible transition to a hierarchical assembly carrying β-sheet-like order, and the dependence of that reversibility on chain length. The two were originally one manuscript and were separated because the dynamical result stands on its own and the structural result belongs with the chain-length series. There is no overlap of figures or

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data between them. We are happy to supply the companion manuscript on request.

Data and code. The reduced data and all analysis and figure-generation code are public at qzhang234/PubCode-2021-IDPP; every data-derived figure regenerates from that repository alone.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing interest; AI-tool use is disclosed in the Acknowledgments. Suggested Editorial Board members, NAS members and reviewers are listed overleaf.

Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D., on behalf of all authors

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Suggested Editors and Reviewers

PNAS asks for a minimum of three Editorial Board members, three NAS members and three qualified reviewers. **TO BE COMPLETED BEFORE SUBMISSION.** The researchers below have relevant expertise in elastin-like polypeptides, protein and peptide assembly, small-angle X-ray scattering, and X-ray photon correlation spectroscopy, and were carried over from the earlier cover letter; they are reviewer suggestions, and the Editorial Board and NAS member lists still have to be chosen.

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